

# Thang K. Tran

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## EDUCATION

University of Massachusetts, Lowell | Lowell, MA

Graduated May 2025

**Bachelor of Science in Business Administration - Management Information Systems**

## EXPERIENCE

**Lightforce Orthodontics** | Process Analyst Co-op

July 2023 - May 2024

- Analyzed production datasets (10k+ records) across 3D printing operations to identify failure patterns and improve ceramic bracket yield
- Conducted data acquisition, cleaning and regression tests with Python for proprietary scanning and processing software and debugged Python modules to resolve 30+ recurring technical issues
- Cut job-ticket turnaround time by ~25% by developing Excel KPI dashboards and working with cross functional teams to execute data-driven workflow improvements
- Performed statistical analysis on resin tensile strength data to determine optimal material mix, reducing print defects by ~15%
- Streamlined 3D printing workflow across IdeaMaker and production tools, ensuring consistent print quality for marketing components under tight production timelines

## PROJECTS

### Census Data Analytics Pipeline - U.S. Education & Income Trends

- Engineered a scalable Python pipeline to ingest and clean nationwide Census data (2015-2020) via the Census API, analyzing education attainment vs household income trends
- Delivered a data storytelling Power BI dashboard for non-technical stakeholders and leadership quantifying regional changes in bachelor's attainment and income, poverty levels, median income and unemployment rates
- Created an ETL pipeline integrating AWS S3, Athena, and Power BI through a relational data model enabling SQL-based analytics and reporting

### Holiday Impact On Retail Sales

- Designed SQL-based ETL pipelines to ingest and clean 6k+ transaction records, engineered star schema for scalable BI reporting in Power BI
- Built a sales analytics dashboard identifying peak revenue months for inventory optimization and demand planning, using a star schema to build data structures in relational form

### Healthcare Insurance Cost Explorer

- Developed a Python-based data visualization tool analyzing 1.3k+ insurance policyholders, revealing smoking status as a 50%+ cost driver and enabling filtering by BMI, region, and gender
- Automated feature correlation and outlier detection in Python (Pandas, NumPy) to support pricing strategy and health risk modeling

## SKILLS

**Languages:** SQL, Python (Pandas/Numpy/Matplotlib/Streamlit/Plotly), DAX, R

**Technologies:** Power BI, Tableau, Power Query, Github, Git, Jira, Slack, Microsoft Excel (Pivot Tables/VLOOKUP), AWS (S3/Athena), Rapidminer, SQL Server Management Studio, Microsoft Office